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LOW-RANK QUADRATIC SPARSE AUTOENCODERS IMPROVE LLM FEATURE RECONSTRUCTION AND INTERPRETABILITY

JOSUÉ CRISTIÁN TAPIA SAN MARTÍN

Thesis submitted to the Office of Research and Graduate Studies
in partial fulfillment of the requirements for the degree of
Magíster en Ciencias de la Ingeniería

Advisor:

JUAN LORENZO REUTTER DE LA MAZA

Santiago de Chile, Abril 2026

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*A Rebeca, Michelle, Christian,
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ABSTRACT

Sparse autoencoders (SAEs) are widely used to decompose large language model (LLM) activations into sparse and interpretable features. Standard linear SAEs, however, impose strong geometric constraints on the representations they can recover, despite growing evidence that many LLM features exhibit intrinsic manifold structure and feature-specific notions of distance rather than behaving as single linear directions.

We introduce a low-rank quadratic sparse autoencoder (LRQ-SAE) that augments the standard linear encoder and decoder with factored quadratic terms, enabling each latent feature to capture controlled second-order interactions while preserving sparsity. Using mid-layer residual-stream activations, we observe that LRQ-SAEs consistently achieve lower normalized reconstruction error than linear SAEs under matched latent dimensionality and sparsity.

To assess interpretability, we employ a robust evaluation protocol in which a larger LLM infers a natural-language definition for each feature from activating examples and then generates new candidate sentences, whose activations are used to estimate feature-level generalization. LRQ-SAEs obtain substantially higher interpretability scores under this metric, indicating that low-rank quadratic structure helps recover more semantically coherent and reliably activated features.

These results suggest that introducing structured second-order capacity relaxes geometric constraints inherent to linear SAEs and shifts the empirical reconstruction–interpretability frontier.

RESUMEN

Los sparse autoencoders (SAE) se utilizan ampliamente para descomponer las activaciones de los modelos de lenguaje en características escasas e interpretables. Sin embargo, los SAE lineales imponen fuertes restricciones geométricas a las representaciones que pueden recuperar, pese a la creciente evidencia de que muchas características en los LLM exhiben estructura de variedad intrínseca y nociones de distancia específicas que no se comportan como direcciones lineales simples.

Presentamos un low-rank quadratic sparse autoencoder (LRQ-SAE) que amplía el codificador y decodificador lineales estándar con términos cuadráticos factorizados, permitiendo que cada característica latente capture interacciones de segundo orden de manera controlada mientras mantiene la escasez. Utilizando activaciones de la corriente residual de capas intermedias, observamos que los LRQ-SAE obtienen de forma consistente menor error de reconstrucción normalizado que los SAE lineales bajo igual dimensionalidad y nivel de dispersión.

Para evaluar la interpretabilidad, empleamos un protocolo robusto en el que un LLM mayor infiere una definición en lenguaje natural para cada característica a partir de ejemplos de activación y luego genera nuevas frases candidatas, cuyas activaciones se utilizan para estimar la generalización a nivel de característica. Los LRQ-SAE obtienen puntuaciones de interpretabilidad sustancialmente más altas bajo esta métrica, lo que indica que la estructura cuadrática de bajo rango ayuda a recuperar características más coherentes semánticamente y activadas de manera más fiable.

Estos resultados sugieren que introducir una capacidad estructurada de segundo orden relaja las restricciones geométricas inherentes a los SAE lineales y desplaza la frontera empírica entre reconstrucción e interpretabilidad.

1. INTRODUCTION

Understanding how large language models (LLMs) represent internal concepts is a central question in mechanistic interpretability. Empirically, an LLM’s hidden state of dimension d can encode far more than d distinct features or concepts. This phenomenon, often called superposition (Elhage et al., 2022), implies that multiple concepts are encoded along overlapping directions rather than each concept having an independent dimension. One theoretical justification is provided by the Johnson-Lindenstrauss lemma, which shows that any set of N points can be embedded into a space of dimension on the order of $\log N$ with little distortion in pairwise distances (Johnson & Lindenstrauss, 1984). Thus a fixed-dimensional representation can preserve the relative structure of exponentially many distinguishable concepts without dedicating an orthogonal axis to each. In other words, an LLM might represent a vast number of features in a single layer’s activations by packing them into nearly orthogonal subspaces. Such an encoding would allow features to be accessed with minimal interference, consistent with the success of linear probes and manipulations on neural representations (Alain & Bengio, 2017), but it also means that these features are entangled in the same vector space and are difficult to interpret directly.

A prominent working model of interpretability is the linear representation hypothesis (LRH), which posits that high-level semantic concepts correspond to approximately linear directions in a model’s activation space. In its simplest form, each concept is associated with a single direction vector, and an activation can be expressed as a sparse linear combination of such feature vectors (Elhage et al., 2022; Park et al., 2024). If true, this idea implies that we can interpret and even control model behaviour using linear algebra, by measuring cosine similarities to known concept vectors or adding and subtracting concept directions to alter representations, leading to a variety of proposed interventions (Alain & Bengio, 2017). Early empirical evidence supported the LRH in simple settings: for example, word-embedding analogies align with linear directions (Mikolov et al., 2013; Pennington et al., 2014), and sparse autoencoders have been applied to test the LRH by

trying to recover these directions from actual LLMs activations (Bricken et al., 2023; Cunningham et al., 2023).

Sparse autoencoders (SAEs) have indeed become a key tool for discovering interpretable features in LLMs. An SAE is an unsupervised model that learns an overcomplete basis in which each input is represented by a sparse code. Intuitively, it finds a set of prototype vectors such that a model activation can be expressed as a combination of only a few active features. This approach, rooted in classic sparse coding and dictionary learning (Arora et al., 2018; Faruqui et al., 2015), has seen a resurgence for neural network interpretability. Recent studies have applied SAEs to various layers of LLMs and other networks, consistently finding that the learned sparse features correspond to human-interpretable concepts (Bricken et al., 2023; Cunningham et al., 2023; Gao et al., 2025). For example, SAE-derived features in GPT-style models have been shown to activate for specific factual or stylistic concepts, such as the presence of certain monuments or the use of polite versus sycophantic phrasing, and analogous techniques in vision models recover features for objects or scenes that align with human semantic categories (Gao et al., 2025).

The interpretability of such features opens the door to useful interventions. If the learned sparse code reflects the actual internal concepts a model uses for its predictions, one can imagine editing representations or ablating features to study causal effects and potentially steer model behaviour. Sparse dictionary learning has therefore been proposed as a way to enumerate the latent concepts an LLM might rely on (Bricken et al., 2023).

Despite their promise, standard SAEs impose strong geometric constraints on the features they can learn, which may limit their ability to capture all types of concepts. Different SAE architectures make different assumptions about how concepts are embedded in the model’s representation space (Hindupur et al., 2025). Many SAEs use a single linear encoder (mapping the d -dimensional activation to a vector of feature activations) followed by a sparsity-inducing nonlinearity (such as a ReLU or top- k threshold) and a linear decoder. This shallow architecture essentially assumes that each concept corresponds to a single fixed direction in $\mathbb{R}^d$ and that features combine linearly. Such an assumption may

be too restrictive for real models. In fact, switching the SAE’s architecture can lead to very different sets of features, suggesting that each SAE is biased toward detecting only certain kinds of features (Hindupur et al., 2025). For example, a ReLU-activated SAE treats each feature as occupying a half-space in activation space, implicitly assuming that concepts are linearly separable by some hyperplane, while a top- k SAE assumes all concepts have the same intrinsic dimensionality and are separated by angle rather than a strict linear boundary (Hindupur et al., 2025). These built-in assumptions mean that if a concept in the model requires a nonlinear combination of neurons or spans a higher-dimensional subspace, a given SAE may fail to discover it (Hindupur et al., 2025). In other words, the SAE does not merely reveal features present in the model; it determines which features can be seen at all. This insight helps explain puzzling observations such as why different sparse coding methods (ℓ_1 regularization versus top- k selection) yield different interpretations even on the same data (Paulo & Belrose, 2025), and it sheds light on instability issues: small changes in the SAE setup can surface a new set of features entirely, indicating that there is no single “universal” SAE that will perfectly extract every concept.

Growing evidence suggests that many LLM features exhibit intrinsic manifold structure. Rather than lying along a single linear axis, a concept’s representation may live on a curved, lower-dimensional surface in the high-dimensional activation space. A feature could therefore have a continuous, multi-dimensional state within the network instead of being simply present or absent. For example, language identity in multilingual models might form a swirling manifold (analogous to a Swiss roll) where one region corresponds to English gradually transitioning into another region for French (Chang et al., 2022). Other work has documented features forming loops or rings, and even tori, in embedding space (Engels et al., 2025; Park et al., 2024). Hierarchical concept relations (like animal taxonomies) can correspond to tree-like structures in representation space (Park et al., 2024). The existence of such structures indicates that the simple linear model of features is incomplete: some concepts are inherently multi-dimensional or vary along a continuum rather than being binary (Engels et al., 2025; Li et al., 2025).

One important implication is that distances in representation space may have feature-specific meaning: small movements along a feature’s manifold can correspond to meaningful semantic variation, while equally sized movements orthogonal to the manifold may not correspond to any coherent state of that concept (Modell et al., 2025). Recent theoretical work shows that if a feature forms a manifold in latent space, cosine similarity between nearby activations can act as a monotonic transformation of the feature’s intrinsic metric, meaning that cosine distance reflects how far two states lie along the shortest path within that manifold (Modell et al., 2025). Taken together, these results suggest that each feature possesses its own local geometry, rather than all features fitting into a single global linear basis.

How can we design sparse interpretable models that accommodate such complex feature geometry? One approach is to modify the SAE architecture to relax its geometric constraints. Hindupur et al., 2025 argued that an optimal sparse autoencoder for a given concept should mirror the structure of that concept in the data. Following this principle, they introduced SpaDE (Sparsemax Distance Encoder), which explicitly builds nonlinear separability and heterogeneous dimensionality into the encoder. Instead of a linear projection followed by a threshold, SpaDE computes distances from the input to a set of learned prototypes and then projects these distances onto a probability simplex via the sparsemax activation (Hindupur et al., 2025; Martins & Astudillo, 2016). This design allows each latent feature to have a flexible, input-dependent receptive field and allows different inputs to activate varying numbers of features (adaptive sparsity). SpaDE can therefore capture concepts that are not linearly separable and those that span varying dimensions, addressing two properties of nonlinear concept structure that earlier SAEs struggled with. Notably, a standard ReLU-based SAE is a special case of SpaDE with particular prototype choices, meaning SpaDE generalizes the linear case to a more flexible metric-based encoding (Hindupur et al., 2025). Beyond SpaDE, other researchers have explored hierarchical sparse autoencoders (Muchane et al., 2025) and route-based SAEs that allow feature-specific sub-networks (Shi et al., 2025), all aiming to better align interpreter architectures with data structure.

In this work we explore an alternative way to relax the geometric constraints imposed by standard sparse autoencoders. Motivated by evidence that many LLM features inhabit curved or multi-dimensional manifolds, we seek a model that expands the representational geometry available to each feature while remaining close to the classical encoder–decoder paradigm. To this end, we introduce a low-rank quadratic sparse autoencoder (LRQ-SAE) that augments the linear formulation with a small set of learned second-order interactions among input dimensions. This structured extension increases the expressiveness of individual features in a controlled manner, enabling them to better align with non-linear regions of activation space without altering the sparsity mechanism or the overcomplete dictionary.

Our central hypothesis is that incorporating low-rank quadratic structure improves an SAE’s ability to capture and disentangle latent features in LLM activations, and simultaneously enhances interpretability. If concepts vary along intrinsically curved trajectories, then allowing features limited second-order capacity should yield more faithful reconstructions and produce activation patterns that are more semantically coherent and consistently triggered across contexts. We evaluate this hypothesis by comparing LRQ-SAEs and linear SAEs under matched sparsity budgets and dictionary sizes, examining how this additional geometric flexibility shifts both the reconstruction–interpretability tradeoff and the stability of feature activation.

2. METHOD

2.1. Data and Activation Collection

We conduct our evaluation on internal activations from a small open-weight language model. Specifically, we use the Pythia-70M model (Biderman et al., 2023), extracting the residual-stream state ($d = 512$) from layer 3 out of 6 for each input token. As our text source, we use the March 2022 snapshot of the Simple English Wikipedia (Wikimedia Foundation, 2022), a corpus of general-domain text.

We sample 10 million tokens from this dataset and record their corresponding residual-stream activations to serve as our training set for the sparse autoencoders. To prevent rare outlier activations from destabilizing training, we apply norm clipping: any activation vector with an ℓ_2 norm above 17.5 is rescaled to have norm 17.5. This retains the direction of extremely large vectors while capping their magnitude, promoting more stable optimization dynamics. The threshold value of 17.5 corresponds approximately to the 95th percentile of the activation-norm distribution, which is where the heavy tail begins to produce extreme outliers. Clipping therefore affects only the upper tail while preserving the typical structure of the residual stream.

2.2. Sparse Autoencoder Architectures and Training

We study a family of low-rank quadratic sparse autoencoders (LRQ-SAEs) obtained by varying the amount of quadratic capacity in the encoder and decoder. All models share the same latent dimensionality $m = 2048$ ($4\times$ the input dimension $d = 512$) and use the same sparsity level $k = 16$ in the BatchTopK operator, enabling controlled comparisons across architectures.

Linear baseline SAE

The configuration with encoder rank $r_{\text{enc}} = 0$ and decoder rank $r_{\text{dec}} = 0$ corresponds to a standard linear sparse autoencoder. The encoder computes the dense preactivation

$$\hat{z}(x) = W_{\text{enc}}x + b_{\text{enc}}, \quad z = \text{BatchTopK}(\hat{z}),$$

and reconstruction is obtained through a linear decoder

$$\hat{x} = W_{\text{dec}}z + b_{\text{dec}}.$$

This model serves as the reference for all comparisons.

Low-rank quadratic encoder

For encoder rank $r_{\text{enc}} \in \{0, 1, 2\}$, we augment the encoder preactivation with a factorized quadratic term:

$$\hat{z}(x) = W_{\text{enc}}x + b_{\text{enc}} + \sum_{i=1}^{r_{\text{enc}}} (U_i x) \odot (V_i x),$$

where $U_i, V_i \in \mathbb{R}^{m \times d}$ and $\odot$ denotes elementwise multiplication. Each factor $(U_i x) \odot (V_i x)$ defines a rank-1 quadratic form in x , so the total quadratic contribution has rank at most r_{enc} . The sparse code is then given by $z = \text{BatchTopK}(\hat{z})$.

Low-rank quadratic decoder

For decoder rank $r_{\text{dec}} \in \{0, 1, 2\}$, the decoder first computes a linear reconstruction

$$x^* = W_{\text{dec}}z + b_{\text{dec}},$$

and then applies a low-rank factorized quadratic correction:

$$\hat{x} = x^* + \sum_{i=1}^{r_{\text{dec}}} (U_i^{\text{dec}} x^*) \odot (V_i^{\text{dec}} x^*),$$

where $U_i^{\text{dec}}, V_i^{\text{dec}} \in \mathbb{R}^{d \times d}$. As in the encoder, each factor contributes a rank-1 quadratic transformation, so the decoder’s quadratic rank is bounded by r_{dec} . When $r_{\text{enc}} = r_{\text{dec}} = 0$, the LRQ-SAE reduces exactly to the linear baseline.

Across the grid $(r_{\text{enc}}, r_{\text{dec}}) \in \{0, 1, 2\}^2$, this construction yields nine LRQ-SAEs that differ only in where and how much quadratic structure is introduced, while keeping latent dimensionality, sparsity, and optimization settings fixed.

Training objective

All models are trained to minimize a normalized reconstruction loss via mean squared error,

$$\text{MSE}(x, \hat{x}) = \frac{\|x - \hat{x}\|^2}{\|x\|^2},$$

augmented with a lightweight statistics-based regularizer on the pre-sparse activations $\hat{z}$. The goal of this regularizer is twofold: (i) to prevent degenerate solutions in which only a small subset of features becomes active across the dataset, and (ii) to ensure gradient flow to all encoder directions, including those that were not selected by BatchTopK in a given batch, allowing inactive features to gradually align with useful directions and avoid dead-feature collapse.

Let μ_j and σ_j denote the mean and standard deviation of feature j within a mini-batch, and let μ_{avg} and σ_{avg} denote their averages across features. The regularizer penalizes deviations from these batch-aggregated statistics:

$$\mathcal{L}_{\text{stats}} = \frac{1}{m} \sum_{j=1}^m (\mu_j - \mu_{\text{avg}})^2 + \frac{1}{m} \sum_{j=1}^m (\sigma_j - \sigma_{\text{avg}})^2.$$

The overall training objective is

$$\mathcal{L} = \text{MSE}(x, \hat{x}) + \lambda \mathcal{L}_{\text{stats}}, \quad \lambda = 0.1.$$

All LRQ-SAEs in the rank grid are trained with the same BatchTopK sparsity level ($k = 16$) and the same AdamW hyperparameters, ensuring that differences in performance can be attributed to encoder-side and decoder-side quadratic structure.

Sparsity via BatchTopK

To enforce sparsity in the codes, we employ the BatchTopK selection method (Bussmann et al., 2024). BatchTopK selects a fixed total number of active features per batch rather than per sample. For each batch, we zero out all but the largest activations across the batch, such that each sample has only a small number of nonzero features on average. This allows flexible allocation: more active features for complex inputs and fewer for simpler ones. We control batch sparsity via a BatchTopK budget k . During inference, we remove the batch dependency and apply a single global threshold τ learned from training. For each feature f we set

$$z_f = \begin{cases} \hat{z}_f, & \text{if } \hat{z}_f > \tau, \\ 0, & \text{otherwise.} \end{cases}$$

Apart from the statistics-based term $\mathcal{L}_{\text{stats}}$, we do not apply additional regularization (e.g., ℓ_1 penalties or orthonormalization of W_{dec}), beyond the implicit regularization of BatchTopK and a small ℓ_2 weight decay.

2.3. Geometric Analysis of Encoder Features

Beyond reconstruction error and sparsity, the LRQ-SAE encoder imposes a specific geometric structure on how each latent feature depends on residual-stream activations. A single feature behaves as a smooth function on the input space whose level sets form hypersurfaces. In the linear SAE these hypersurfaces are all parallel and flat, while in the LRQ-SAE they acquire controlled curvature determined directly by the encoder parameters. This section formalizes this structure and introduces a simple curvature metric that we later relate to reconstruction and interpretability.

Single-feature encoder and active domain

Let $x \in \mathbb{R}^d$ denote a residual-stream activation and let $f \in \{1, \dots, m\}$ index a latent feature. For an LRQ encoder with rank $r_{\text{enc}} = r$, the preactivation of feature f is

$$\hat{z}_f(x) = w_f^\top x + \sum_{i=1}^r (\langle u_{i,f}, x \rangle) (\langle v_{i,f}, x \rangle) + b_f,$$

where $w_f \in \mathbb{R}^d$ is the f -th row of W_{enc} , $b_f \in \mathbb{R}$ is the corresponding bias, and $u_{i,f}, v_{i,f} \in \mathbb{R}^d$ are the f -th rows (viewed as column vectors) of the factor matrices $U_i, V_i \in \mathbb{R}^{m \times d}$.

During inference, the global threshold τ described in the BatchTopK subsection selects the active domain

$$X_f := \{x \in \mathbb{R}^d : \hat{z}_f(x) > \tau\},$$

on which $z_f(x) = \hat{z}_f(x)$. All geometric quantities are computed using $\hat{z}_f$ restricted to X_f .

Level sets as hypersurfaces in residual-stream space

For any value $c > \tau$, the level set of feature f is

$$S_{f,c} := \{x \in X_f : z_f(x) = c\}.$$

Each $S_{f,c}$ is a hypersurface embedded in $\mathbb{R}^d$. In the linear baseline ($r = 0$),

$$z_f(x) = w_f^\top x + b_f,$$

so all level sets are parallel affine hyperplanes. They do not curve or change orientation across space: the feature behaves as a fixed linear direction.

When $r > 0$, the quadratic terms allow the level sets to bend and rotate, forming curved hypersurfaces (e.g., bowls, saddles, or mixtures thereof). Importantly, this curvature is confined to a low-dimensional subspace determined by the learned vectors $\{u_{i,f}, v_{i,f}\}_{i=1}^r$, which means the feature can express richer geometric structure without incurring the full complexity of a general quadratic map.

Hessian and curvature of encoder features

The curvature of these hypersurfaces is governed entirely by the Hessian of the function z_f ,

$$H_f := \nabla^2 z_f(x).$$

The affine term contributes zero curvature, while each quadratic factor contributes a symmetric rank-2 matrix. A direct calculation shows that

$$\nabla^2 (\langle u_{i,f}, x \rangle \langle v_{i,f}, x \rangle) = u_{i,f} v_{i,f}^\top + v_{i,f} u_{i,f}^\top,$$

so the Hessian of feature f takes the closed form

$$H_f = \sum_{i=1}^r (u_{i,f} v_{i,f}^\top + v_{i,f} u_{i,f}^\top).$$

This matrix is independent of x and has rank at most $2r$. Thus, each rank-1 quadratic factor contributes curvature in at most two independent directions, giving each feature a “curvature budget” proportional to $2r$. The linear SAE is recovered as the special case $H_f = 0$.

Feature-specific metric and collapse of level sets

A single feature induces a feature-specific metric on the active domain X_f by

$$d_f(x, y) := |z_f(x) - z_f(y)|, \quad x, y \in X_f.$$

Formally, d_f is a pseudometric (distinct points can have zero distance), but it is natural to view (X_f, d_f) as a metric space from the perspective of feature f . All points lying on the same level set $S_{f,c}$ satisfy $d_f(x, y) = 0$ and are therefore collapsed to a single equivalence class, even when they are far apart in Euclidean space. Conversely, small Euclidean displacements that cross level sets can yield large values of d_f , reflecting a substantial change in the feature.

This viewpoint aligns with (Modell et al., 2025), where each feature is treated as a metric space endowed with its own feature-specific notion of distance. We use d_f purely as an interpretive device and do not rely on it in our quantitative evaluation.

Curvature metrics for encoder features

Because H_f is fully determined by the encoder parameters of feature f , its magnitude provides a direct measure of how much curvature the LRQ-SAE assigns to that feature. We summarize curvature with the Frobenius norm

$$\kappa_f^{\text{Frob}} := \|H_f\|_F,$$

which measures the overall strength of second-order structure. In the linear SAE this metric is exactly zero for all features. For LRQ-SAEs with $r > 0$ it can be non-zero, and in our trained models it is in fact bounded away from zero for almost all features. Comparing the distribution of $\{\kappa_f^{\text{Frob}}\}$ across architectures reveals how encoder-side quadratic structure is actually used in practice and how much curvature each model allocates to its learned features.

2.4. Evaluation Metrics and Procedure

We evaluate reconstruction fidelity, feature interpretability, and encoder-side curvature on a random sample of features.

For reconstruction, we report the normalized mean squared error (MSE) used in the training objective. To assess how interpretable the learned features are, we employ an activation-based concept inference procedure in which a strong external LLM, in this case Gemini 2.5 (Comanici et al., 2025), acts as an independent evaluator. Rather than relying on manual inspection or cluster-based heuristics, this approach tests whether the behaviour of each feature can be explained and predicted by another model.

For each autoencoder, we begin by sampling 500 non-dead features. For a given feature f , we uniformly sample ten text contexts from the same corpus among those positions where the feature is active, that is, where $z_f > \tau$. Each snippet is paired with its corresponding activation magnitude. Because these examples are selected randomly rather than by peak activation, they provide a more diverse and representative view of the contexts in which the feature fires. These activation–context pairs are supplied to Gemini 2.5, which infers a concise natural-language hypothesis describing the underlying concept.

Once Gemini proposes a hypothesis, we request five new passages that, according to its inferred concept, should activate the same feature. These candidate triggers are processed through the Pythia-70M network to obtain residual-stream activations, which are then encoded by the autoencoder. For each passage we record whether the resulting sparse code reactivates f , i.e., whether $z_f > \tau$. The interpretability score of feature f is defined as the fraction of successful activations out of the five attempts, yielding a continuous value in $[0, 1]$ that quantifies how reliably an external model can elicit feature activation from newly constructed inputs.

After repeating this procedure across the sampled features, we compute the mean interpretability score for each LRQ-SAE configuration. To quantify differences across models, we perform a two-sample t -test comparing each configuration against the linear baseline with $r_{\text{enc}} = r_{\text{dec}} = 0$. This establishes whether encoder-side or decoder-side quadratic structure yields statistically significant improvement in feature interpretability.

In addition, our evaluation departs from the standard procedure used in recent work (O’Neill et al., 2025; Paulo et al., 2025) in two key ways. First, the external LLM does not assign the interpretability score itself; instead, the score is determined solely by whether the autoencoder reactivates the feature on the LLM-generated candidate passages. Second, rather than selecting only top-activation examples, we draw a random sample from all positions where the feature is active. This yields a more diverse and representative set of contexts, reducing bias toward extreme or prototypical activations.

3. RESULTS

3.1. Reconstruction performance

We first compare LRQ-SAEs and the linear baseline in terms of reconstruction error and parameter count. Table 3.1 reports the number of trainable parameters as a function of encoder and decoder rank. Introducing quadratic structure roughly doubles the parameter count when moving from $r_{\text{enc}} = 0$ to $r_{\text{enc}} = 1$, and yields an overall increase of about a factor of three for the largest model ($r_{\text{enc}}=2, r_{\text{dec}}=2$). Even in this highest-capacity configuration, however, the total parameter count remains on the order of a few million and is still negligible compared with the 70M parameters of the underlying language model.

Figure 3.1 summarizes normalized reconstruction error as a function of the BatchTopK budget k . As expected, increasing k systematically lowers mean squared error for all architectures, since denser codes allow the decoder to use more features per input. More importantly, for every value of $k \in \{8, 10, 12, 14, 16\}$, all LRQ-SAEs with non-zero encoder rank achieve consistently lower MSE than the linear baseline ($r_{\text{enc}}=0, r_{\text{dec}}=0$). The gain is already visible for $r_{\text{enc}}=1$ and becomes more pronounced for $r_{\text{enc}}=2$. At $k = 16$, the best quadratic model ($r_{\text{enc}}=2, r_{\text{dec}}=2$) reduces reconstruction error by roughly 10–15% relative to the baseline, despite being trained under identical sparsity and optimization settings. Decoder-side quadratic structure provides an additional but smaller improvement: for fixed encoder rank, increasing r_{dec} from 0 to 2 yields a modest yet consistent reduction in MSE, with diminishing returns at higher ranks.

To relate reconstruction quality to interpretability, Figure 3.2 plots MSE against the Feature Interpretability Score (FIS) for all nine configurations at $k = 16$. The points follow an approximate monotone trend: architectures that reconstruct better also tend to achieve higher FIS. In particular, the ($r_{\text{enc}}=2, r_{\text{dec}}=1$) and ($r_{\text{enc}}=2, r_{\text{dec}}=2$) models lie in the top-left corner of the plot, simultaneously minimizing MSE and maximizing FIS. This empirical alignment indicates that, in this regime, adding low-rank quadratic capacity does

Table 3.1. Parameter count (in millions) for LRQ-SAE models as a function of encoder and decoder rank. Percentages indicate increase relative to the baseline ($r_{\text{enc}}=0, r_{\text{dec}}=0$).

r_{enc}	r_{dec}		
	0	1	2
0	2.1M	2.6M (+25%)	3.2M (+50%)
1	4.2M (+100%)	4.7M (+125%)	5.2M (+150%)
2	6.3M (+200%)	6.8M (+225%)	7.3M (+250%)

Table 3.2. Feature Interpretability Score (FIS) for all LRQ-SAE configurations at $k = 16$. Each cell reports FIS with the activation threshold τ in parentheses. Higher FIS indicates more reliably interpretable features.

r_{enc}	r_{dec}		
	0	1	2
0	0.432 (0.710)	0.449 (0.502)	0.448 (0.397)
1	0.523 (0.716)	0.526 (0.632)	0.519 (0.526)
2	0.543 (0.729)	0.574 (0.654)	0.548 (0.570)

Table 3.3. Welch t-tests comparing FIS against the linear baseline ($r_{\text{enc}}=0, r_{\text{dec}}=0$) at $k = 16$. Negative t-statistics indicate higher FIS than the baseline.

r_{enc}	r_{dec}	t-statistic	p-value
0	0	-	-
0	1	-0.8154	4.15×10^{-1}
0	2	-0.7774	4.37×10^{-1}
1	0	-4.3272	1.66×10^{-5}
1	1	-4.4170	1.11×10^{-5}
1	2	-4.0409	5.73×10^{-5}
2	0	-5.2341	2.02×10^{-7}
2	1	-6.5899	7.12×10^{-11}
2	2	-5.3703	9.79×10^{-8}

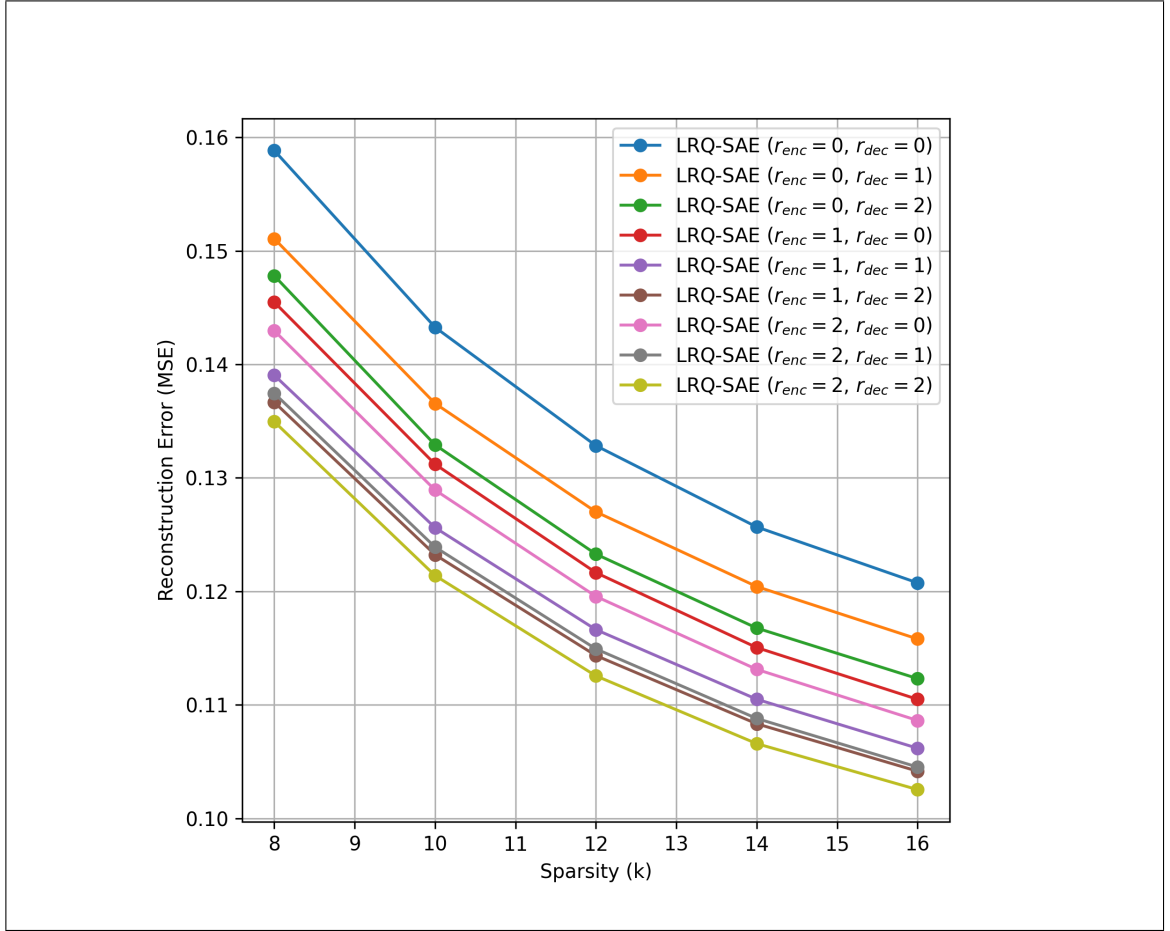


Figure 3.1. Reconstruction error (MSE) as a function of the sparsity budget k for all LRQ-SAE configurations. Increasing k systematically reduces MSE across all models, as expected from denser latent codes. For every value of k , configurations with higher encoder rank achieve lower reconstruction error, with $(r_{enc}=2, r_{dec}=2)$ consistently obtaining the lowest MSE.

not trade off reconstruction for interpretability; instead, both quantities improve together when encoder rank is increased.

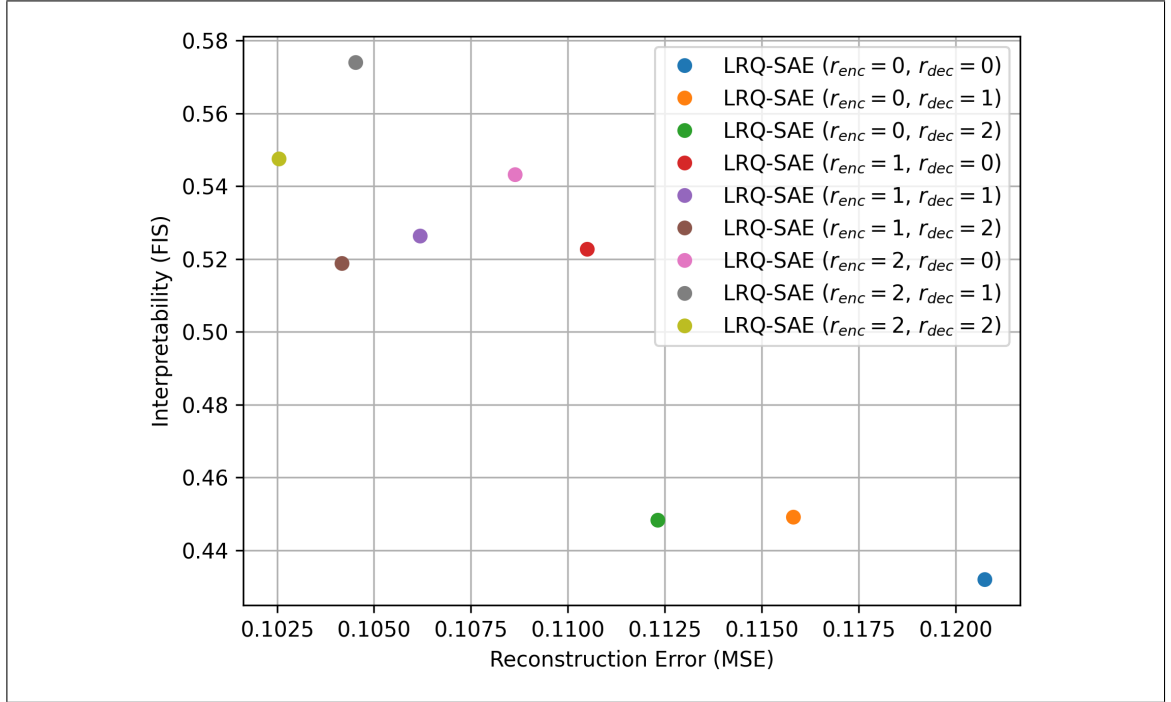


Figure 3.2. Reconstruction error (MSE) versus interpretability (FIS) for all LRQ-SAE configurations at $k = 16$. Each point corresponds to a distinct encoder–decoder rank pair. Configurations with higher encoder rank achieve both lower MSE and higher FIS, with $(r_{enc}=2, r_{dec}=1)$ attaining the best combined performance.

3.2. Feature interpretability

Interpretability was measured using the activation-based concept inference test described in the Methods section. Table 3.2 reports the mean FIS for each LRQ-SAE, together with the activation threshold τ estimated from training. The linear baseline achieves a mean FIS of 0.432, meaning that, on average, fewer than half of the hypotheses produced by the external evaluator successfully re-activate the corresponding feature on new examples. Introducing quadratic structure in the encoder yields substantial gains. All configurations with $r_{enc} > 0$ obtain FIS values in the range 0.52–0.57, representing an absolute improvement of approximately 0.09–0.14 over the baseline and a relative increase of around 20–30%. The best-performing model ($r_{enc} = 2, r_{dec} = 1$) attains $FIS = 0.574$,

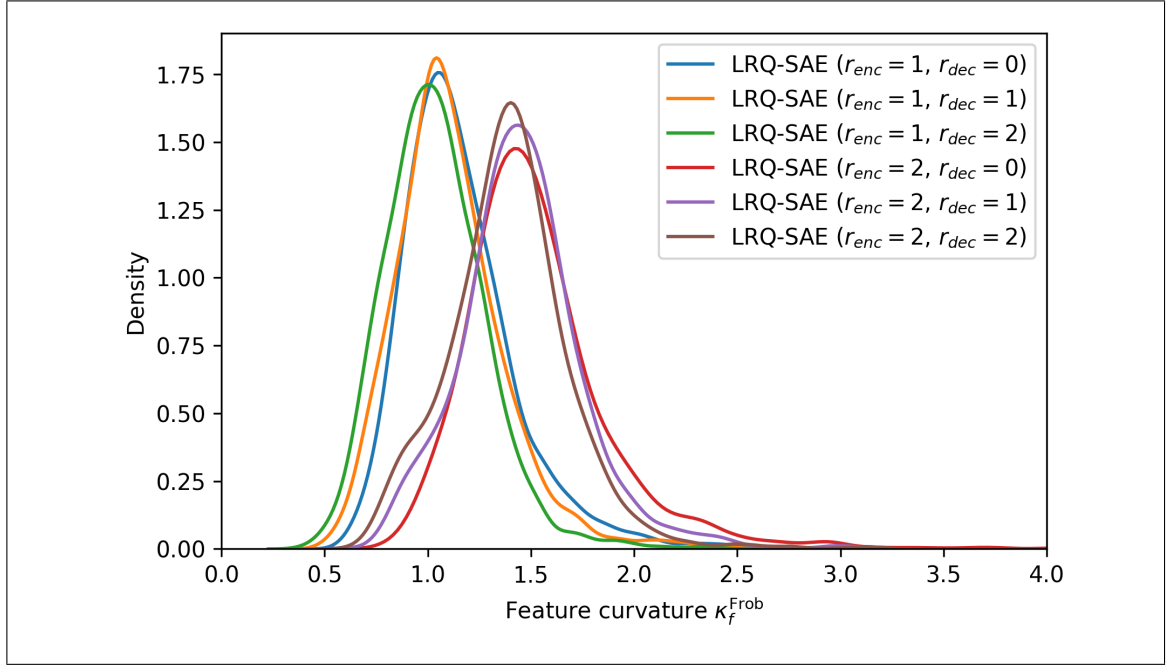


Figure 3.3. Kernel density estimates of feature curvature κ_f^{Frob} for LRQ-SAE models with $r_{\text{enc}} \in \{1, 2\}$ and $r_{\text{dec}} \in \{0, 1, 2\}$. Increasing encoder rank shifts the curvature distribution toward higher values, indicating that quadratic encoder structure produces features whose local manifolds exhibit stronger nonlinear variation. Decoder rank introduces additional but smaller shifts in curvature, suggesting that encoder-side nonlinearity is the primary contributor to geometric complexity in the learned features.

indicating that, for most features, the evaluator can reliably elicit activations on the majority of its candidate triggers.

The activation threshold τ also follows a consistent pattern across architectures. Thresholds tend to increase with encoder rank. This suggests that quadratic encoders learn features that are both more selective and more semantically reliable when they do activate, as reflected in higher FIS. Decoder rank, in contrast, exerts only a weak effect on FIS when r_{enc} is held fixed, reinforcing the observation that encoder-side structure is the main determinant of interpretability in this setting.

Table 3.3 reports Welch t -tests comparing the FIS distributions of each LRQ-SAE against the linear baseline. For models with $r_{\text{enc}} = 0$ but $r_{\text{dec}} > 0$, the differences are not

statistically significant ($p \approx 0.4$), indicating that decoder-side quadratic structure alone does not meaningfully change feature interpretability under our metric. By contrast, all configurations with $r_{\text{enc}} > 0$ show highly significant improvements, with p -values well below 10^{-4} and often below 10^{-7} . The largest effect is observed for $(r_{\text{enc}} = 2, r_{\text{dec}} = 1)$, whose FIS distribution differs from the baseline with $t = -6.59$ and $p \approx 7 \times 10^{-11}$. Taken together, these results support that increasing encoder rank systematically enhances the semantic consistency and predictability of the learned features, whereas decoder rank plays a secondary role.

3.3. Encoder curvature and feature geometry

We now relate these empirical gains to the geometric diagnostics introduced in the Methods section. Figure 3.3 reports kernel density estimates of the Frobenius-norm curvature κ_f^{Frob} for all non-dead features across LRQ-SAEs with encoder ranks $r_{\text{enc}} \in \{1, 2\}$ and decoder ranks $r_{\text{dec}} \in \{0, 1, 2\}$. The linear SAE baseline is omitted because its Hessian is identically zero for all features.

Across all models with $r_{\text{enc}} = 1$, curvature values cluster tightly around a well-defined mode near $\kappa_f^{\text{Frob}} \approx 1$, with no mass near zero. This indicates that even the lowest-rank LRQ encoder learns features that exhibit clear second-order structure. Although most features fall within a narrow band, the distributions show a visible tail extending toward higher curvature values, reflecting localized departures from linearity in a small number of directions.

Increasing encoder rank from 1 to 2 uniformly shifts the distributions to the right and broadens them. The modes move toward approximately $\kappa_f^{\text{Frob}} \approx 1.5$, and the upper tails extend further past 2. Since the Hessian rank cannot exceed $2r_{\text{enc}}$, this shift reflects the enlarged quadratic subspace available to rank-2 encoders. In practice, the learned models exploit this capacity, producing features whose activation regions bend along more independent directions in the residual-stream space.

Variations in decoder rank produce only secondary effects. Within a fixed encoder rank, the curves for $r_{\text{dec}} = 0, 1, 2$ remain closely aligned, differing mostly by small horizontal displacements rather than qualitative changes in shape. This is consistent with the architectural separation between encoder geometry and decoder reconstruction: curvature is introduced analytically through the encoder’s quadratic terms, while the decoder acts only on the sparse code z and therefore does not modify the structure of the encoder level sets that determine feature activation.

Taken together, these results support a coherent geometric interpretation. Allowing higher encoder rank yields features with systematically larger and more variable curvature, suggesting that LRQ-SAEs use their quadratic capacity to fit genuinely curved activation regions rather than near-linear ones. These curved features are also the ones associated with improved reconstruction quality and higher FIS, indicating that controlled second-order geometry provides a more faithful alignment with the intrinsic manifolds present in mid-layer LLM activations.

4. DISCUSSION

Our experiments provide evidence that introducing low-rank quadratic structure into sparse autoencoders can improve both reconstruction fidelity and interpretability of LLM internal representations. Across a controlled grid of encoder and decoder ranks, LRQ-SAEs with non-zero encoder rank consistently outperform the linear baseline in normalized MSE (Figure 3.1), while also achieving substantially higher Feature Interpretability Scores (Table 3.2). These gains are obtained under matched sparsity, latent dimensionality, and optimization settings. In the joint MSE vs. FIS plot of Figure 3.2, the configurations approximately follow a trend in which increasing encoder rank moves models toward both lower reconstruction error and higher FIS; the $(r_{\text{enc}}=2, r_{\text{dec}}=1)$ model attains the best combined values among all nine configurations.

From a geometric perspective, the LRQ encoder can be viewed as endowing each feature with a low-rank curvature budget in residual-stream space. Theoretical analysis shows that the Hessian of the feature preactivation has rank at most $2r_{\text{enc}}$ and is fully determined by the pairs of encoder vectors $\{u_{i,f}, v_{i,f}\}$. Empirically, increasing encoder rank shifts the distribution of curvature κ_f^{Frob} toward larger values and broadens its tail (Figure 3.3), confirming that LRQ-SAEs actually use this additional capacity to realize genuinely curved level sets rather than remaining close to linear. These curved features are not arbitrary: they correlate with both lower reconstruction error and higher FIS, indicating that the quadratic encoder is capturing meaningful nonlinear structure in the residual stream rather than overfitting to noise or idiosyncratic artefacts.

The alignment between reconstruction and interpretability is particularly notable. In many settings, additional representational capacity can be spent on modeling nuisance variation that improves MSE while degrading semantic coherence, or vice versa. Here, however, architectures with higher encoder rank tend to move toward better values of both MSE and FIS (Figure 3.2), suggesting that the linear SAE is underfitting the true geometry of the representation space. Its hard bias toward flat hyperplanes prevents it from aligning

with curved manifolds along which concepts vary, as suggested by recent work on representation manifolds in LLMs (Modell et al., 2025). In practice, attempting to capture a highly curved concept with a linear SAE may force the dictionary to approximate the manifold with several overlapping linear segments, leading to feature-splitting behaviour closely related to the splitting phenomena studied by Chanin et al. (2025). Allowing features a controlled amount of curvature via low-rank quadratic terms appears to relax this bias just enough for the sparse dictionary to better track the intrinsic manifolds of LLM features, without sacrificing the interpretability benefits of sparsity and overcompleteness.

Our results also clarify the asymmetric roles of encoder and decoder structure. Decoder-side quadratic corrections provide modest gains in reconstruction but do not significantly change interpretability when encoder rank is held at zero (Tables 3.2 and 3.3). By contrast, even small encoder ranks ($r_{\text{enc}} = 1$) yield clear improvements in FIS, and $r_{\text{enc}} = 2$ further boosts both reconstruction and interpretability. This asymmetry is conceptually natural: the encoder determines which regions of residual-stream space activate a feature and thus controls the shape and selectivity of the feature’s manifold, while the decoder maps the resulting sparse code back to activations without influencing when a feature fires. If interpretability is fundamentally about how features carve up representation space, it is reasonable that the encoder’s geometry plays the dominant role.

A natural alternative way to accommodate curved feature geometry is to replace the linear encoder with a small multi-layer perceptron. This approach has recently been explored by O’Neill et al. (2025), who study MLP encoders for SAEs and report improved interpretability over a linear baseline under their evaluation protocol. However, their interpretability metric differs from the one used here in two important respects: they focus on top-activation examples, and they ask the external LLM to assign an interpretability score to each feature. In our FIS protocol, by contrast, activations are sampled uniformly

from all active positions and the score is determined solely by whether the autoencoder re-activates the feature on LLM-generated candidate passages. In additional preliminary experiments, applying our FIS protocol to an MLP encoder improved reconstruction but degraded interpretability relative to the linear SAE baseline ($MSE = 0.116$, $FIS = 0.395$). A plausible explanation is that an unconstrained MLP encoder has too much freedom: it can easily entangle multiple concepts into a single feature or implement highly irregular activation regions that are difficult to summarize semantically. The LRQ encoder occupies a middle ground between a purely linear map and a generic MLP, adding only structured low-rank quadratic interactions whose geometric effects are analytically tractable. This combination of controlled expressivity and explicit geometric structure may explain why LRQ-SAEs can realize curved manifolds while still producing features that remain amenable to human and LLM-based interpretation.

The activation-based interpretability metric we use, FIS, offers a complementary perspective to purely reconstruction-based or clustering-based evaluations. By asking an external evaluator to infer and then test a natural-language hypothesis for each feature, FIS measures whether feature activations can be predicted from a coherent semantic description rather than merely whether they are sparse or statistically disentangled. The fact that LRQ-SAEs achieve higher FIS indicates that their features are not only more geometrically aligned with the data but also more amenable to semantic explanation and controlled activation. At the same time, the divergence between our FIS-based findings and the MLP results reported by O’Neill et al. (2025) underscores that interpretability conclusions can depend sensitively on the evaluation protocol, reinforcing the need for multiple, conceptually grounded metrics.

4.1. Limitations and future work

The present study has several limitations. First, all experiments are conducted on a single base model (Pythia-70M), a single layer (residual stream at layer 3), and a single corpus (Simple English Wikipedia). It remains to be seen whether the same patterns

hold for larger LLMs, deeper or earlier layers, or domain-specific corpora. For instance, representations closer to the input or output may exhibit different manifold structure and sparsity patterns, potentially changing the relative benefits of quadratic encoders.

Second, we restrict attention to very low quadratic ranks ($r_{\text{enc}}, r_{\text{dec}} \in \{0, 1, 2\}$) in order to keep parameter counts modest and curvature analysis tractable. Higher ranks could unlock richer geometries, allowing features to bend along more directions and support more complex manifolds, but may also risk overfitting or destabilizing training. Systematically exploring this regime, including regularization strategies or architectural constraints that keep higher-rank encoders well behaved, is an important direction for future work.

Third, our curvature metric is global and encoder-centric: κ_f^{Frob} summarizes second-order structure using a single scalar computed from the Hessian and does not explicitly account for where in input space a feature tends to activate. More refined analyses could distinguish principal curvature directions, relate curvature to specific semantic dimensions, or combine encoder-side geometry with the empirical distribution of activations to characterize which parts of a feature’s manifold are actually used by the base model. Similarly, our interpretability metric relies on a particular external evaluator (Gemini 2.5) and a specific prompt design. Although the FIS procedure is in principle applicable with human annotators, we have only tested it with a single LLM and prompt. Alternative evaluators, prompt schemes, and task designs could provide a more robust picture of how LRQ-SAEs affect interpretability.

Fourth, our comparison with MLP encoders remains preliminary. While the evidence here suggests that unconstrained MLP encoders can degrade interpretability under the FIS protocol, a more systematic study is needed to disentangle the effects of architecture, capacity, and evaluation methodology. Directly comparing LRQ-SAEs, linear SAEs, and MLP-based SAEs across multiple interpretability metrics, including both top-activation-based scores and activation-driven tests such as FIS, would help clarify when additional nonlinear capacity helps or hurts interpretability, and why.

Finally, we do not attempt to intervene on the base model using LRQ-SAE features. Our results show that these features are more semantically coherent and reliably activatable, but we do not yet know whether editing or ablating them produces more targeted and predictable changes in model behaviour than edits based on linear SAEs. Establishing such causal efficacy, for example by systematically manipulating LRQ-SAE features and measuring downstream effects on generations, would substantially strengthen the case for adopting LRQ-SAEs as practical interpretability tools for model steering, auditing, and safety.

5. CONCLUSION

We have introduced low-rank quadratic sparse autoencoders as a simple extension of linear SAEs that relaxes their geometric constraints while preserving sparsity and over-completeness. By augmenting the encoder and decoder with factored quadratic terms, LRQ-SAEs allow each feature to depend on residual-stream activations through a low-rank quadratic form rather than a single linear direction. On the mid-layer residual stream of Pythia-70M, this additional second-order capacity yields consistent improvements in normalized reconstruction error and a concept-based Feature Interpretability Score, with the strongest gains arising from quadratic structure in the encoder.

From a geometric point of view, our results support the view that many LLM features behave as low-dimensional curved manifolds in representation space rather than as single linear directions. The low-rank Hessians induced by LRQ encoders provide just enough curvature to better align the sparse dictionary with these manifolds, in a way that is both quantitatively beneficial (lower MSE, higher FIS) and qualitatively interpretable. More broadly, this suggests that the architecture of the sparse autoencoder should be treated as an explicit hypothesis about feature geometry, and that incorporating controlled nonlinear structure can be a productive way to bridge the gap between linear SAEs and the intrinsic representations learned by large language models.

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APPENDIX

A. PROMPT INSTRUCTIONS AND QUALITATIVE FEATURE ANALYSIS

This appendix provides supplementary material for the interpretability evaluation of sparse autoencoders (SAEs) trained on residual-stream activations of a language model. We include: (i) the exact prompt template used during LLM-based feature evaluation (§A.1), and (ii) representative qualitative analyses for two human-interpretable features, one from the linear baseline SAE and one from the LRQ-SAE (§A.2). Feature interpretability scores (FIS) are computed as the fraction of positive examples correctly matched by the LLM.

For all FIS values and full feature explanations across all models, the complete set of interpretations can be consulted at <https://github.com/josuetism/lrq-sae>.

A.1. LLM Prompt Template Used for Feature Evaluation

The following text corresponds exactly to the prompt shown to the LLM evaluator when producing explanations and proposing new activating sentences for each feature.

You are analyzing a single SAE (Sparse Autoencoder) feature in a
 ↪ language model. Activated tokens are marked as <<token>>(score).
 ↪ Ignore the markers when writing NEW sentences.

TASKS:

- 1) Under "explanation": briefly describe the common trigger or
 ↪ pattern.
- 2) Under "activating_sentences": provide EXACTLY 5 NEW sentences that
 ↪ SHOULD strongly activate this feature.

GUIDANCE:

- Avoid reusing the exact same trigger tokens or phrases as in the
 ↪ examples.
- You may vary topic, wording, or structure as long as the sentences
 ↪ are likely to activate the feature.

OUTPUT (STRICT):

- Return ONLY a valid JSON object with EXACTLY these keys:
 ↪ "explanation", "activating_sentences".
- "activating_sentences": array of 5 strings.
- Valid JSON only (double quotes, no trailing commas). No code
 ↪ fences, no comments, no extra text.

OUTPUT EXPECTATION:

A single valid JSON object with the two keys above and nothing else.

A.2. Representative Qualitative Examples for Two SAE Models

We present qualitative analyses for two features:

- Feature 766 from the linear baseline SAE ($r_{\text{enc}} = 0$, $r_{\text{dec}} = 0$), which consistently fires on mentions of death in biographical contexts.

- Feature 834 from the LRQ-SAE ($r_{\text{enc}} = 2$, $r_{\text{dec}} = 1$), which captures temporal spans and references to historical periods.

For each feature we show:

- the LLM-generated explanation,
- a subset of positive activation examples drawn directly from the evaluation dataset,
- the five new activating sentences generated by the LLM,
- the empirical FIS (Feature Interpretability Score).

Linear baseline SAE ($r_{\text{enc}} = 0$, $r_{\text{dec}} = 0$)

Feature 766. FIS = 1.0 (5/5).

LLM-generated explanation.

This feature activates on mentions of death, particularly in
 → biographical contexts. It triggers when a person's death is
 → stated or implied, often followed by details such as the cause of
 → death, specific date, location, or age at the time of death. It
 → also frequently activates on specific dates and years, especially
 → when associated with life events like births or deaths, and
 → sometimes on the "d. YYYY" annotation.

Positive examples shown to the evaluator (subset).

Example 1:

ilova, Russian ballerina (d. 1997)
 1904 - John MacCormick, Scottish lawyer and politician (d.
 → 1961)<<\n>>(0.80) 1908 - Alistair Cooke, British-born American
 → journalist (d. 2004)
 1911 - Jean Shiley, American high jumper (d. ...)

Example 2:

democracy.

On 1 January 2012, she was<< honored>>(0.77) by being place on
→ Malawis 200 Kwacha banknote.

Chibambo<< died>>(1.99)<< from>>(2.59)<< a>>(3.04)<< heart>>(2.80)<<
→ attack>>(3.96)<< at>>(3.22)<< a>>(3.23)<< hospital>>(3.16)<<
→ in>>(2.79)<< Bl>>(1.00)anty<<re>>(1.92)<<, >>(2.70)<<
→ Mal>>(0.76)<<awi>>(1.80)<< at>>(2.33)<< the>>(1.99)<<
→ age>>(1.86)<< of>>(1.33)<< 87>>(2.12)<<.>>(2.51)

<<\n>>(0.76)References

1928 births

2016 ...

Example 3:

On 28 March 2020, Schfer was found<< dead>>(1.17)<< at>>(1.83)<<
→ a>>(1.84)<< railway>>(1.76)<< line>>(1.69)<< near>>(0.97)
→ Hoch<<heim>>(0.84)<< am>>(1.39)<< Main>>(1.34)<<.>>(1.37)
→ Police<< ruled>>(0.77)<< his>>(0.86)<< death>>(1.15)<<
→ a>>(1.21)<< suicide>>(1.18)<<.>>(0.91) He was<<
→ 54>>(1.17)<<.>>(0.90)

References

1966 births

2020 deaths

Politicians from Hesse

Politicians of the Christian Democratic Union of Germany ...

LLM-proposed activating sentences.

The renowned author passed away peacefully due to natural causes,
→ having reached 92 years.
She breathed her last breath October 23, 2005, residing in her
→ cherished countryside home.

Professor Eleanor Vance (b. 1930, obit. 2011) was a pioneer in
→ neuroscience.

He succumbed to a lengthy illness in 2019, being 68 years when he
→ departed.

The tragic demise of the explorer occurred during an expedition to
→ the arctic wilderness, December 1st, 1988.

LRQ-SAE ($r_{\text{enc}} = 2, r_{\text{dec}} = 1$)

Feature 834. FIS = 1.0 (5/5).

LLM-generated explanation.

This feature activates for phrases and words indicating chronological
→ spans, durations, and specific points within historical or
→ temporal sequences. It is particularly sensitive to references to
→ decades, eras, and words denoting a beginning, end, or
→ progression through time.

Positive examples shown to the evaluator (subset).

Example 1:

ocaster, Precision Bass, and the Jazz Bass) became popular. Fender
→ Electric Instrument Manufacturing Company sold thousands and
→ thousands of these instruments. Many rock and pop music bands
→ from<< the>>(1.54)<< 1960>>(2.18)<<s>>(1.35)<< to>>(1.01)<<
→ the>>(1.69)<< 2000>>(2.35)<<s>>(1.49) use these electric guitars
→ and electric bass guitars.

Example 2:

ism took hold from 1924 on, the state began limiting the freedom of
→ artists. From<< the>>(0.72)<< late>>(4.90)<<
→ 1920>>(1.06)<<s>>(1.53) the Russian avant-garde experienced
→ direct and harsh criticism from the authorities and in 1934 the
→ doctrine of socialist realism became official policy, and
→ prohibited abstraction and different artistic expression.

Example 3:

Family from the<< late>>(4.29)<< 1960>>(2.45)<<s>>(0.88) but<<
→ the>>(0.78) episodes are not put on television. Mike Evans plays
→ Lionel in All in the Family and the first season<< of>>(0.72) The
→ Jeffersons. Damon Evans plays Lionel after Mike Evans left but
→ Mike came back to play the character in 1979.

LLM-proposed activating sentences.

Between 1970 and 1985, global temperatures showed a steady increase.
This particular epoch, extending from 1450 A.D. through 1750 A.D.,
→ brought profound changes to Europe.
Construction work, beginning in May and concluding in December, faced
→ numerous delays.
Numerous scientific advancements happened throughout early
→ twenty-first century, continuing into its middle segment.
From its inception in 2005 onwards, an organization expanded its
→ global reach considerably.